

Three Phase MMC modelling including multiple controls

Sounak Nandi

January 14, 2024

1 Introduction

The MMC in this work is modelled on the concept of averaged theory. The modelling takes into account of complex frequency cross-coupling among different state variables which is accomplished by linearization along a steady state solution.

1.1 Nomenclature

$j=u,l$

$k=a,b,c$

u_{kg} and i_{kg} =output voltage and current in the AC side

U_g and I_g =amplitude of output voltage and current

f_1 =Fundamental Frequency

V_{dc} and i_{dc} =DC-side voltage and Current

P =rated power

Φ =power factor angle

C =submodule capacitance

C_{arm} =equivalent capacitance of arms

u_{kj} and i_{kj} =arm voltage and current

i_{kc} =circulating current

L_a =arm inductor

m_1 =modulation ratio

N =submodule number per arm

R_a =equivalent arm resistance

u_{kcj}^i =capacitor voltage of the i^{th} submodule

$u_{kcj} = \sum_{i=1}^N u_{kcj}^i$ =total capacitor voltage of arms

$u_{kc\Delta}$ =difference capacitor voltage between arms

$u_{kc\Sigma}$ =total capacitor voltage of phases

S_{kj} =Switching Functions of Arms

$S_{k\Delta} = S_{ku} - S_{kl}$ =Differential-mode component of a switching function

$S_{k\Sigma} = S_{ku} + S_{kl}$ =Common-mode component of a switching function

T_s =Sampling Period

1.2 Computation of the Steady state components

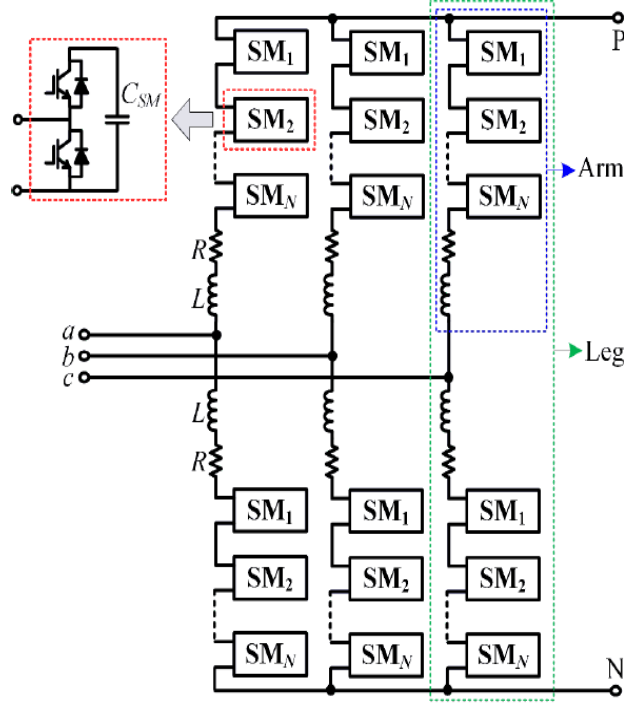


Figure 1: Typical Circuit Diagram of MMC[From 3]

2. The state space equations are derived based on the following Figures 1 and

$$\frac{di_{kc}}{dt} = -\frac{R_a}{L_a} i_{kc} - \frac{S_k \sum^t}{4L_a} u_{kc \Sigma} + \frac{S_k \Delta t}{4L_a} u_{kc \Delta} + \frac{1}{2L_a} u_{kg} \quad (1)$$

$$\frac{di_{kg}}{dt} = -\frac{R_a}{L_a}i_{kg} - \frac{S_k\Delta t}{2L_a}u_{kc}\Sigma - \frac{S_k\Sigma t}{2L_a}u_{kc\Delta} - \frac{2}{L_a}V_{dc} \quad (2)$$

$$\frac{du_{kc}\Sigma}{dt} = \frac{S_{k\Sigma t}}{C_{arm}}i_{kc} + \frac{S_{k\Delta t}}{2C_{arm}}i_{kg} \quad (3)$$

$$\frac{du_{kc\Delta}}{dt} = \frac{S_{k\Delta t}}{C_{arm}} i_{kc} + \frac{S_k \sum t}{2C_{arm}} i_{kg} \quad (4)$$

The A matrix on per phase basis will be:

$$A(t) = \begin{pmatrix} -\frac{R_a}{L_a} & 0 & -\frac{S_k \sum t}{4L_a} & \frac{S_k \Delta t}{4L_a} \\ 0 & -\frac{R_a}{L_a} & -\frac{S_k \Delta t}{2L_a} & -\frac{S_k \sum t}{2L_a} \\ \frac{S_k \sum t}{C_{arm}} & \frac{S_k \Delta t}{2C_{arm}} & 0 & 0 \\ \frac{S_k \Delta t}{C_{arm}} & \frac{S_k \sum t}{2C_{arm}} & 0 & 0 \end{pmatrix} \quad (5)$$

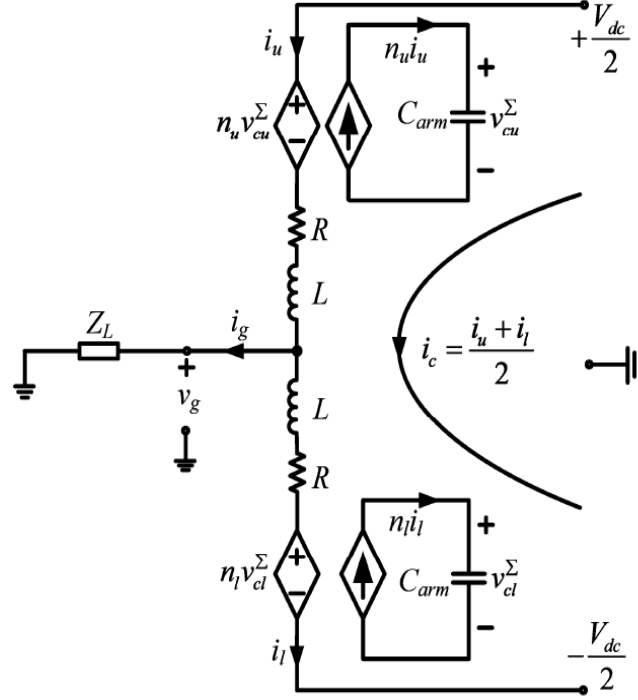


Figure 2: Averaged Value Model of One-Phase of a MMC[From 3]

The B matrix is :

$$B(t) = \begin{pmatrix} 0 & \frac{1}{2L_a} \\ -\frac{2}{L_a} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \quad (6)$$

The states are:

$$x(t) = [i_{kc}, i_{kg}, u_{kc\Sigma}, u_{kc\Delta}] \quad (7)$$

The input matrix is:

$$u = [V_{dc}, u_{kg}] \quad (8)$$

The steady state switching functions of the arms on per phase(say phase a) is usually composed of DC, fundamental and second harmonic components which can be expressed as:

$$S_{ku} = \frac{m_0}{2} - \frac{m_1 \sin(\omega_1 t + \theta_1)}{2} - \frac{m_2 \sin(\omega_2 t + \theta_2)}{2} \quad (9)$$

$$S_{kl} = \frac{m_0}{2} + \frac{m_1 \sin(\omega_1 t + \theta_1)}{2} - \frac{m_2 \sin(\omega_2 t + \theta_2)}{2} \quad (10)$$

1.2.1 Harmonic State Space Modelling

The state equations for a Linear time Periodic System is :

$$sX = (A - N)X + BU \quad (11)$$

$$Y = CX + DU \quad (12)$$

Now A, B, C and D are LTP equivalent matrices of $A(t), B(t), C(t)$ and $D(t)$. The matrix $N = \text{diag}(jm\omega_o)$. Getting the Fourier Coefficients of a time-varying signal

The signal $x(t)$ needs to be decomposed into complex Fourier series of the form:

$$x(t) = \sum_{-\infty}^{+\infty} X_n \exp(jn\omega_o t) \quad (13)$$

So, $x(t)$ can be written in the form,

$$x(t) = \Gamma(t)X \quad (14)$$

where

$$\Gamma(t) = [\exp(-jh\omega_o t), \exp(-j(h-1)\omega_o t), \dots, 1, \dots, \exp(j(h-1)\omega_o t), \exp(jh\omega_o t)] \quad (15)$$

$$X = [X_{-h}(t), X_{-(h-1)}(t), \dots, X_0(t), \dots, X_{(h-1)}(t), X_h(t)]^T \quad (16)$$

Note: The above series is truncated up to the h harmonic.

Toeplitz Matrix In LTP domain, while performing convolution Toeplitz Transformation is required. The Toeplitz Transformation of a vector of the form

$$SW = [SW_{-h}, \dots, SW_0, \dots, SW_h] \quad (17)$$

is

$$\Gamma[SW] = \begin{bmatrix} SW_0 & SW_{-1} & SW_{-2} & \dots \\ SW_1 & SW_0 & SW_{-1} & \dots \\ SW_2 & SW_1 & SW_0 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \quad (18)$$

Defining the parameters to be used in the code:

$P = 96 * 10^6$; Rated Power
 $f = 50$; Rated Frequency
 $w1 = 2 * \pi * f$;
 $V_{dc} = 40 * 10^3$; Rated DC Voltage
 $V_{ac} = 16 * 10^3$; Rated AC Voltage
 $N = 100$; submodule number per arm
 $C_{sm} = 20 * 10^{-6}$; submodule capacitor
 $C_{arm} = C_{sm}/N$;
 $L = 5 * 10^{-3}$; arm inductance

$R = 0.1$; arm resistance

1.3 Construction of Toeplitz Vectors for the switching functions

Defining the switching functions

$m_0=0.9944; m_1=0.8094; \theta_1=0.1354; m_2=0.0302; \theta_2=1.6686;$

$ps1p=\exp(((4*\pi)/3)*i); ps2p=\exp(((2*\pi)/3)*i);$ [phase shift for the b and c phases from a phase for the fundamental frequency]

$ps1n=\exp(-1*((4*\pi)/3)*i); ps2n=\exp(-1*((2*\pi)/3)*i);$ [phase shift for the b and c phase for the second harmonic]

Defining common-mode component of switching functions

Phase a

$fac1n=(m_2/(2*i))*\exp(-1*\theta_2*i); fac1p=(m_2/(2*i))*\exp(1*\theta_2*i);$

$vcma=[0;-1*fac1n;0;m_0;0;fac1p;0];$ [vector for the toeplitz matrix of a phase switching]

Phase b

$vcmb=[0;-1*fac1n*ps1p;0;m_0;0;fac1p*ps1n;0];$ [vector for the toeplitz matrix of b phase switching]

Phase c

$vcmc=[0;-1*fac1n*ps1n;0;m_0;0;fac1p*ps1p;0];$ [vector for the toeplitz matrix of c phase switching]

Defining differential-mode component of switching functions

Phase a

$fac2n=(m_1/(2*i))*\exp(-1*\theta_1*i); fac2p=(m_1/(2*i))*\exp(1*\theta_1*i);$

$vdma=[0;0;fac2n;0;-1*fac2p;0;0];$ [vector for the toeplitz matrix of a phase switching]

Phase b

$vdmb=[0;0;fac2n*ps2p;0;-1*fac2p*ps2n;0;0];$ [vector for the toeplitz matrix of b phase switching]

Phase c

$vdmc=[0;0;fac2n*ps2n;0;-1*fac2p*ps2p;0;0];$ [vector for the toeplitz matrix of c phase switching]

For Instance:

$$\Gamma[vcma] = \begin{bmatrix} m_0 & -fac1n & 0 & 0 & 0 & 0 & 0 \\ fac1p & m_0 & -fac1n & 0 & 0 & 0 & 0 \\ 0 & fac1p & m_0 & -fac1n & 0 & 0 & 0 \\ 0 & 0 & fac1p & m_0 & -fac1n & 0 & 0 \\ 0 & 0 & 0 & fac1p & m_0 & -fac1n & 0 \\ 0 & 0 & 0 & 0 & fac1p & m_0 & -fac1n \\ 0 & 0 & 0 & 0 & 0 & fac1p & m_0 \end{bmatrix} \quad (19)$$

1.4 Computation of the steady state point

The steady-state response of the state variables is given by $X_{ss} = -A^{-1}BU$.

Now the state variable response obtained from the equation is only the Fourier Coefficients. To convert them back in the time domain, we need an operator $P(t)$. $x(t) = P(t)X$, where $P(t) = [\exp(-ih\omega_1 t), \exp(-i(h-1)\omega_1 t), \dots, 1, \dots, \exp(i(h-1)\omega_1 t), \exp(ih\omega_1 t)]$ and $X^T = [X_{-h}, X_{-(h-1)}, \dots, X_0, \dots, X_{h-1}, X_h]$. We obtain the X vector from the steady-state calculations.

In Matlab, the function `timedomain.m` is written to perform the conversion of a vector into a time domain.

The Output from the steady state analysis is shown below.

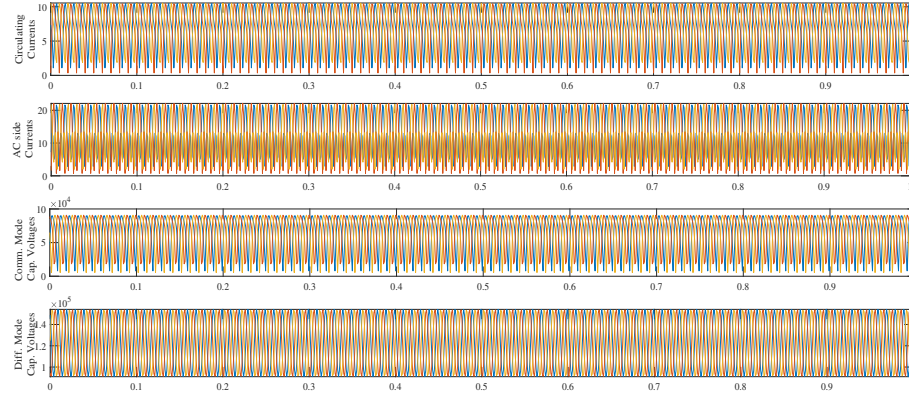


Figure 3: Steady State Output

2 Closed Loop Control Implementation

2.1 Modeling the Topology Part

The topology part is linearized around the steady state values.

$$\frac{d\Delta x_p}{dt} = A_p(t)\Delta x_p(t) + B_p(t)\Delta u_p(t) \quad (20)$$

where,

$$\Delta x_p(t) = [\Delta i_{kc}(t), \Delta i_{kg}(t), \Delta u_{kc\sum}(t), \Delta u_{kc\Delta}(t)] \quad (21)$$

and,

$$\Delta u_p(t) = [\Delta S_{k\sum}, \Delta S_{k\Delta}, \Delta V_{dc}, \Delta u_{kg}] \quad (22)$$

$$A_p(t) = \begin{pmatrix} -\frac{R_a}{L_a} & 0 & -\frac{S_k \Sigma t}{4L_a} & \frac{S_k \Delta t}{4L_a} \\ 0 & -\frac{R_a}{L_a} & -\frac{S_k \Delta t}{2L_a} & -\frac{S_k \Sigma t}{2L_a} \\ \frac{S_k \Sigma t}{C_{arm}} & \frac{S_k \Delta t}{2C_{arm}} & 0 & 0 \\ \frac{S_k \Delta t}{C_{arm}} & \frac{S_k \Sigma t}{2C_{arm}} & 0 & 0 \end{pmatrix} \quad (23)$$

$$B_p(t) = \begin{pmatrix} -\frac{u_{kc} \Sigma s}{4L_a} & \frac{u_{kc} \Delta s}{4L_a} & \frac{1}{2L_a} & 0 \\ -\frac{u_{kc} \Delta s}{2L_a} & -\frac{u_{kc} \Sigma s}{2L_a} & 0 & -\frac{2}{L_a} \\ \frac{i_{kcs}}{C_{arm}} & \frac{i_{kgs}}{2C_{arm}} & 0 & 0 \\ \frac{i_{kgs}}{2C_{arm}} & \frac{i_{kcs}}{C_{arm}} & 0 & 0 \end{pmatrix} \quad (24)$$

2.2 Modeling of the Control Part

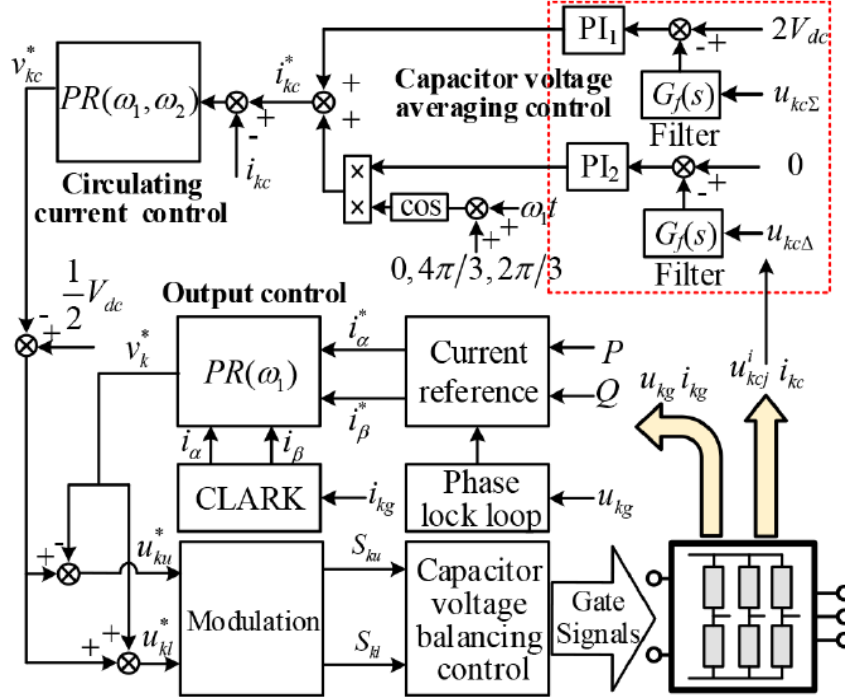


Figure 4: Detailed Control Structure[4]

The moving average filter tuned at the fundamental frequency is adopted as the capacitor voltage filter $G_f(s)$.

$$G_f(s) = \frac{1 - e^{-sT_1}}{sT_1} \quad (25)$$

which can be approximated as,

$$G_f(s) = \frac{1}{1 + (1/2)T_1s + (1/12)T_1^2s^2} \quad (26)$$

So the model of the moving average filter in the control loop is described by:

$$\frac{d\Delta u_{kc\Sigma_f}}{dt} = \Delta x_1 \quad (27)$$

$$\frac{d\Delta x_1}{dt} = -\frac{12}{T_1^2}\Delta u_{kc\Sigma_f} - \frac{6}{T_1}\Delta x_1 + \frac{12}{T_1^2}\Delta u_{kc\Sigma} \quad (28)$$

$$\frac{d\Delta u_{kc\Delta_f}}{dt} = \Delta x_2 \quad (29)$$

$$\frac{d\Delta x_2}{dt} = -\frac{12}{T_1^2}\Delta u_{kc\Delta_f} - \frac{6}{T_1}\Delta x_2 + \frac{12}{T_1^2}\Delta u_{kc\Delta} \quad (30)$$

where, $\Delta u_{kc\Sigma_f}$ and $\Delta u_{kc\Delta_f}$ are filtered capacitor voltages of the upper and lower arm.

Note: The DC and the fundamental component of the circulating currents are well regulated so that capacitor voltage averaging is achieved.

The resulted perturbation in the circulating current is:

$$\Delta i_{kc}^*(t) = K_{pv1}\Delta u_{kc\Sigma_f}(t) + K_{pv2}\Delta u_{kc\Delta_f}(t)\cos\omega_1 t + K_{iv1}\Delta x_3(t) + K_{iv2}\Delta x_4(t)\cos\omega_1 t \quad (31)$$

$$\frac{d\Delta x_3(t)}{dt} = -\Delta u_{kc\Sigma_f}(t) \quad (32)$$

$$\frac{d\Delta x_4(t)}{dt} = \Delta u_{kc\Delta_f}(t) \quad (33)$$

Note: The resonant part of the PR controller only affects the controlled variable around the designated frequency while having little impact on the other frequencies.

The perturbations of the arm voltage references are obtained as:

$$\begin{aligned} \Delta u_{ku}^*(t) &= K_{pc}\Delta i_{kc}(t) - K_{po}\Delta i_{kg}(t) - K_{pc}K_{iv1}\Delta x_3(t) \\ &- K_{pc}K_{iv2}\Delta x_4(t)\cos\omega_1 t + K_{pc}K_{pv1}\Delta u_{kc\Sigma_f}(t) - K_{pc}K_{pv2}\Delta u_{kc\Delta_f}(t)\cos\omega_1 t \end{aligned}$$

$$\begin{aligned} \Delta u_{kl}^*(t) &= K_{pc}\Delta i_{kc}(t) + K_{po}\Delta i_{kg}(t) - K_{pc}K_{iv1}\Delta x_3(t) \\ &- K_{pc}K_{iv2}\Delta x_4(t)\cos\omega_1 t + K_{pc}K_{pv1}\Delta u_{kc\Sigma_f}(t) - K_{pc}K_{pv2}\Delta u_{kc\Delta_f}(t)\cos\omega_1 t \end{aligned}$$

$$G_d(s) = e^{-sT_s} \frac{1 - e^{-sT_s}}{sT_s} \simeq e^{-1.5sT_s} \quad (34)$$

$$G_d(s) = \frac{1 - (3/4)sT_s}{1 + (3/4)sT_s} \quad (35)$$

$$\Delta S_{ku}(t) = \Delta x_5(t) - \frac{\Delta u_{ku}^*(t)}{V_{dc}} \quad (36)$$

$$\Delta S_{kl}(t) = \Delta x_6(t) - \frac{\Delta u_{kl}^*(t)}{V_{dc}} \quad (37)$$

$$\frac{d\Delta x_5(t)}{dt} = -\frac{4}{3T_s} \Delta x_5(t) + \frac{8}{3T_s} \frac{\Delta u_{ku}^*(t)}{V_{dc}} \quad (38)$$

$$\frac{d\Delta x_6(t)}{dt} = -\frac{4}{3T_s} \Delta x_6(t) + \frac{8}{3T_s} \frac{\Delta u_{kl}^*(t)}{V_{dc}} \quad (39)$$

$$A_c(t) = \begin{pmatrix} -\frac{6}{T_1} & 0 & 0 & 0 & 0 & 0 & -\frac{12}{T_1^2} & 0 \\ 0 & -\frac{6}{T_1} & 0 & 0 & 0 & 0 & 0 & -\frac{12}{T_1^2} \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -\frac{8K_{pc}}{3T_s V_{dc}} K_{pc} K_{iv1} - \frac{8K_{pc}}{3T_s V_{dc}} K_{iv2} \cos(\omega_1 t) - \frac{4}{3T_s} & 0 & \frac{8K_{pc}}{3T_s V_{dc}} K_{pv1} - \frac{8K_{pc}}{3T_s V_{dc}} K_{pv2} \cos(\omega_1 t) & & & \\ 0 & 0 & -\frac{8K_{pc}}{3T_s V_{dc}} K_{pc} K_{iv1} - \frac{8K_{pc}}{3T_s V_{dc}} K_{iv2} \cos(\omega_1 t) & 0 & -\frac{4}{3T_s} \frac{8K_{pc}}{3T_s V_{dc}} K_{pv1} - \frac{8K_{pc}}{3T_s V_{dc}} K_{pv2} \cos(\omega_1 t) & & & \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (40)$$

$$B_c(t) = \text{transpose} \begin{pmatrix} 0 & 0 & 0 & 0 & \frac{8K_{pc}}{3T_s V_{dc}} & \frac{8K_{pc}}{3T_s V_{dc}} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{8K_{pc}}{3T_s V_{dc}} & \frac{8K_{pc}}{3T_s V_{dc}} & 0 & 0 \\ \frac{12}{T_1^2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{12}{T_1^2} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (41)$$

$$C_c(t) = \begin{pmatrix} 0 & 0 & \frac{2K_{pc}}{V_{dc}} K_{iv1} & \frac{2K_{pc} K_{iv2}}{V_{dc}} \cos(\omega_1 t) & 1 & 1 & -\frac{2K_{pc}}{V_{dc}} K_{pv1} & \frac{2K_{pc}}{V_{dc}} K_{pv2} \cos(\omega_1 t) \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \end{pmatrix} \quad (42)$$

$$D_c(t) = \begin{pmatrix} -\frac{2K_{pc}}{V_{dc}} & 0 & 0 & 0 \\ 0 & \frac{2K_{pc}}{V_{dc}} & 0 & 0 \end{pmatrix} \quad (43)$$

$$\Delta x_c(t) = [\Delta x_1(t), \Delta x_2(t), \Delta x_3(t), \Delta x_4(t), \Delta x_5(t), \Delta x_6(t), \Delta u_{kc \Sigma_f}(t), \Delta u_{kc \Delta_f}(t)] \quad (44)$$

$$\Delta y_c(t) = [\Delta S_{k \Sigma}(t), \Delta S_{k \Delta}(t)] \quad (45)$$

$$\Delta u_c(t) = [\Delta i_{kc}(t), \Delta i_{kg}(t), \Delta u_{kc \Sigma}(t), \Delta u_{kc \Delta}(t)] \quad (46)$$

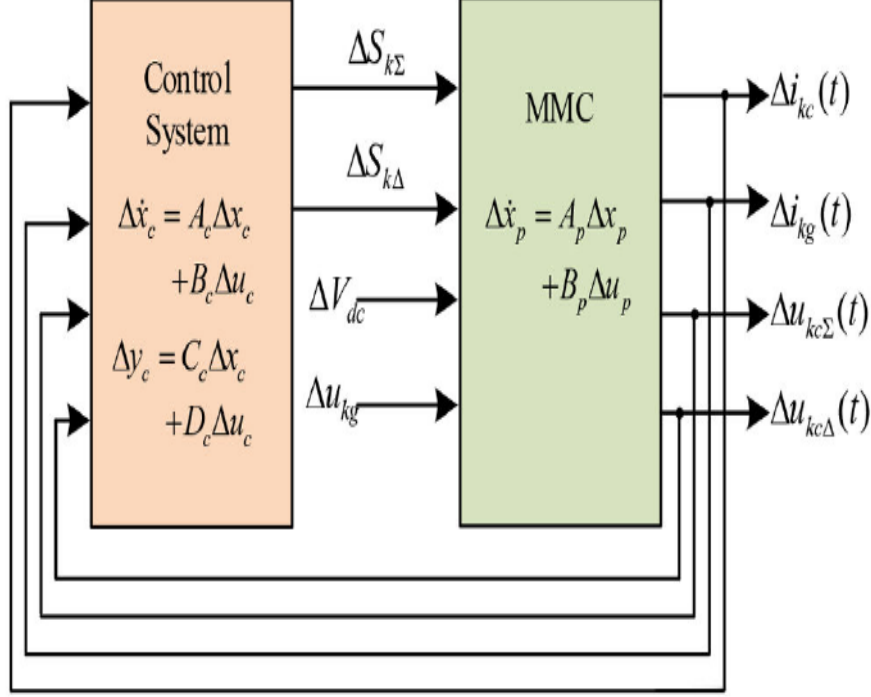


Figure 5: Combined Topology and Control[4]

3 Combining Topology and Control

$$\alpha(t) = \begin{pmatrix} A_c(t) & B_c(t) \\ B_{p1}(t)C_c(t) & A_p(t) + B_{p1}(t)D_c(t) \end{pmatrix} \quad (47)$$

$$\beta(t) = \begin{pmatrix} 0 \\ B_{p2}(t) \end{pmatrix} \quad (48)$$

$$\left[\frac{d\Delta x_c(t)}{dt}, \frac{d\Delta x_p(t)}{dt} \right] = \begin{pmatrix} \alpha(t)[\Delta x_c(t), \Delta x_p(t)] \\ \beta(t)\Delta u_{p2}(t) \end{pmatrix} \quad (49)$$

$$s\Delta X = (A - N)\Delta X + B\Delta U \quad (50)$$

Note: $B_{p1}(t)$ is the first two columns of $B_p(t)$ and $B_{p2}(t)$ is the last two columns of the B_p and u_{p1} and u_{p2} are the first and last rows of u_p .

4 References

- 1 J. Lyu, X. Zhang, X. Cai and M. Molinas, "Harmonic State-Space Based Small-Signal Impedance Modeling of a Modular Multilevel Converter With Consideration of Internal Harmonic Dynamics," in IEEE Transactions

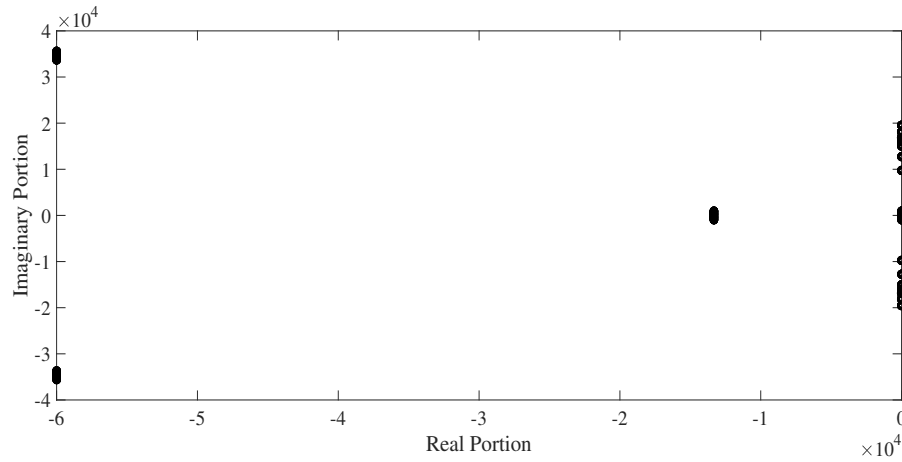


Figure 6: Eigenvalues of the Closed Loop System

on Power Electronics, vol. 34, no. 3, pp. 2134-2148, March 2019, doi: 10.1109/TPEL.2018.2842682.

- 2 J. Lyu, X. Cai, X. Zhang and M. Molinas, "Harmonic State Space Modeling and Analysis of Modular Multilevel Converter," 2018 IEEE International Power Electronics and Application Conference and Exposition (PEAC), Shenzhen, China, 2018, pp. 1-6, doi: 10.1109/PEAC.2018.8590305.
- 3 Jing Lyu, Marta Molinas, Xu Cai, "Harmonic State Space Modeling of a Three-Phase Modular Multilevel Converter", [https://arxiv.org/ftp/arxiv/papers/1706/1706.09925.pdf].
- 4 Yajun Ma, Hua Lin, Zhe Wang, Zuyao Ze, "Stability analysis of modular multilevel converter based on harmonic state-space theory", IET Power Electronics, 2019, Vol. 12, Iss. 15, pp. 3987-3997.